Rate-Limit Forecast Model

claumon `internal/forecast`

This document specifies the math behind `internal/forecast`. It is normative: implementations must match the formulas here, and deviations are bugs unless this document is updated.

1 What we forecast

For each open rate-limit window (session, weekly, per-model weekly), the forecaster reads the time series of utilization snapshots and produces:

1. A point estimate F of the utilization at reset.
2. An 80% **credible interval** $[F^-, F^+]$ for F. (We use the looser "confidence" wording in user-facing strings only; the posterior is Bayesian, see §5.)
3. For each threshold C_{thr} (e.g. 100%), either a median ETA with an 80% CI, or `nil` if the threshold is unlikely to be reached before reset.
4. A confidence tag in $\{\text{Low}, \text{Medium}, \text{High}\}$.

This procedure is an *empirical Bayes* forecast: the prior on the rate r and the path-noise variance σ^2 are both estimated from past data (§5, §7) and then plugged into the posterior update, rather than being themselves marginalized under a hyperprior. The conjugate update on r and the Monte Carlo over rate-and-path uncertainty are genuinely Bayesian; the "empirical" qualifier refers to the prior, not the inference step.

A separate forecast is produced per gauge. The only cross-window sharing is through the calibration constants (§7).

2 The model in one paragraph

Inside the current window, utilization $u(t)$ grows at an unknown constant rate r plus Brownian noise (random behavioral pivots). We estimate r from the recent slope of the snapshots (OLS) and from a Gaussian prior fit on past sessions, combined by conjugate update. The forecast at reset is the sum of two random pieces: deterministic-time-times-uncertain-rate, plus Brownian accumulation over the remaining horizon. Both pieces are Gaussian, so the forecast is Gaussian and the CI is a z-quantile. The ETA has no closed form, so we Monte Carlo it.

3 Generative model and assumptions

For $s \in [0, \Delta t_{\text{rem}}]$ with $\Delta t_{\text{rem}} = T_{\text{reset}} - t_{\text{now}}$,

$$u(t_{\text{now}} + s) = u_{\text{now}} + r\, s + W(s), \qquad W(s) \sim \mathcal{N}\big(0,\, \sigma^2_{\text{session}}\, s\big),$$

with r unknown and W a Brownian motion independent of r. Three assumptions:

1. **Linear conditional mean.** r is constant within a window. Violated by abrupt mid-window mode shifts; partially absorbed by the recency window in §5.

2. **Brownian path noise.** Variance grows linearly with time. Approximately correct if behavioral pivots are mean-zero and not too heavy-tailed.

3. **Exchangeable past sessions.** Historical sessions are iid draws from the same noise process. Required for the calibration in §7.

4 Notation

Symbol	Meaning
$u(t)$, u_{now}	utilization at time t and at now, $\in [0,1]$
t_{now}, T_{reset}, Δt_{rem}	now, reset, remaining horizon (h)
r	true mean growth rate within window (h^{-1})
$\hat{r}_{\text{OLS}}$, SE_{OLS}	OLS estimate and its standard error
μ_0, τ_0^2	Gaussian prior on r from history
$\hat{r}_{\text{post}}$, τ^2_{post}	posterior on r after the OLS update
$\sigma^2_{\text{session}}$	Brownian diffusion coefficient (h^{-1})
F, σ_F^2	point forecast and its variance
$z_{0.9}$	$\Phi^{-1}(0.9) \approx 1.2816$
K, Δs	MC trajectories, step size (defaults: 500, 5 min)

5 Estimating the rate

Combine two sources of information about r.

From the current window. Let $\mathcal{R}$ be the snapshots in $[t_{\text{now}} - \tau_{\text{recent}}, t_{\text{now}}]$ and $n = |\mathcal{R}|$. For $n \geq 3$, fit $u_i = \alpha + r\, t_i + \epsilon_i$ by OLS:

$$\hat{r}_{\text{OLS}} = \frac{S_{tu}}{S_{tt}}, \qquad \text{SE}^2_{\text{OLS}} = \frac{\hat{\sigma}^2_\epsilon}{S_{tt}},$$

with $S_{tt} = \sum(t_i - \bar{t})^2$, $S_{tu} = \sum(t_i - \bar{t})(u_i - \bar{u})$, and $\hat{\sigma}^2_\epsilon = \frac{1}{n-2}\sum(u_i - \hat{\alpha} - \hat{r}_{\text{OLS}}\, t_i)^2$. Default $\tau_{\text{recent}} = 30\,\text{min}$.

Strictly, Brownian residuals are heteroskedastic and serially correlated, so the iid-OLS variance formula above is an approximation. For the short recency window the discrepancy is small and the point estimate is unbiased either way; we accept the approximation and treat SE^2_{OLS} as the likelihood precision in the conjugate update below.

From history. For each past completed session s with final value u_s^* and duration D_s, define $\rho_s = u_s^*/D_s$. The prior mean is the sample mean of ρ_s. For the prior variance, note that under the generative model

$$\rho_s = r_s + \frac{W_s(D_s)}{D_s}, \qquad \text{Var}[\rho_s] = \text{Var}[r_s] + \frac{\sigma^2_{\text{session}}}{D_s},$$

so the raw sample variance of ρ_s overstates $\text{Var}[r_s]$ by the average path-noise contribution. Subtract it off:

$$\mu_0 = \text{mean}_s\, \rho_s, \qquad \tau_0^2 = \max\Big(\text{var}_s\, \rho_s - \sigma^2_{\text{session}} \cdot \text{mean}_s(1/D_s),\ \varepsilon\Big),$$

with $\varepsilon = 10^{-6}$ guarding against negative estimates from small samples. This uses the most recent $\sigma^2_{\text{session}}$ from §7; the two fits depend on each other but only loosely, and the daily refit cycle converges quickly. On the very first fit (before §7 has run), $\sigma^2_{\text{session}} = 0$ and the correction is a no-op.

Fit at startup, refresh daily.

Combine. Normal-normal conjugacy gives the posterior $r \mid \hat{r}_{\mathrm{OLS}} \sim \mathcal{N}(\hat{r}_{\mathrm{post}}, \tau^2_{\mathrm{post}})$:

$$\frac{1}{\tau^2_{\mathrm{post}}} = \frac{1}{\tau_0^2} + \frac{1}{\mathrm{SE}^2_{\mathrm{OLS}}}, \qquad \hat{r}_{\mathrm{post}} = \tau^2_{\mathrm{post}} \left(\frac{\mu_0}{\tau_0^2} + \frac{\hat{r}_{\mathrm{OLS}}}{\mathrm{SE}^2_{\mathrm{OLS}}} \right).$$

Limits: prior dominates when $\mathrm{SE}_{\mathrm{OLS}}$ is large (early window); data dominates when $\mathrm{SE}_{\mathrm{OLS}}$ is small (later in the window). For $n < 3$ the OLS step is undefined; use $\hat{r}_{\mathrm{post}} = \mu_0$, $\tau^2_{\mathrm{post}} = \tau_0^2$.

6 Forecast distribution at reset

The point forecast is

$$F = u_{\mathrm{now}} + \hat{r}_{\mathrm{post}}\,\Delta t_{\mathrm{rem}}.$$

By the law of total variance, conditioning on r:

$$\sigma_F^2 = \underbrace{\Delta t^2_{\mathrm{rem}}\,\tau^2_{\mathrm{post}}}_{\text{rate uncertainty}} + \underbrace{\Delta t_{\mathrm{rem}}\,\sigma^2_{\mathrm{session}}}_{\text{path noise}}.$$

Both contributions are Gaussian under the model, so $u(T_{\mathrm{reset}}) \sim \mathcal{N}(F, \sigma_F^2)$. The two terms have different scaling: rate uncertainty is quadratic in horizon (dominates at long horizons), path noise is linear (dominates at short).

80% CI. $[F - z_{0.9}\,\sigma_F,\ F + z_{0.9}\,\sigma_F]$, clipped to $[0,1]$ for display. The unclipped F is retained for ETA computation.

7 Calibration of $\sigma^2_{\mathrm{session}}$

What we learn. One scalar: how much utilization can drift from the trend per hour of waiting. It is the only quantity history contributes to the forecast *spread* (the prior μ_0, τ_0^2 contributes to the *mean*).

Strategy. Run the forecaster on past sessions, where we already know the answer, and measure how wrong it was as a function of horizon. Read $\sigma^2_{\mathrm{session}}$ off the empirical error structure.

Replay. For each past session s with reset value u_s^* at T_s, sample forecast points $t_f \in [t_s + \tau_{\mathrm{recent}}, T_s - \Delta_{\mathrm{min}}]$ (default 6 per session, $\Delta_{\mathrm{min}} = 30\,\mathrm{min}$). At each t_f, run §5 on snapshots in $[t_f - \tau_{\mathrm{recent}}, t_f]$ to obtain $\hat{r}_{\mathrm{post}}(t_f)$, and the projection $\hat{F}(t_f) = u(t_f) + \hat{r}_{\mathrm{post}}(t_f)\,(T_s - t_f)$. Record

$$e_f = u_s^* - \hat{F}(t_f), \qquad \delta_f = T_s - t_f.$$

Two sources of error in each residual. The residual e_f mixes (A) the rate estimate being off at t_f, with the error compounded over δ_f, and (B) Brownian path noise accumulating over δ_f. By the same decomposition as §5,

$$\mathbb{E}[e_f^2 \mid \delta_f] = \underbrace{\delta_f\,\sigma^2_{\mathrm{session}}}_{\text{path noise (B)}} + \underbrace{\delta_f^2\,\bar{\tau}^2}_{\text{rate uncertainty (A)}},$$

with $\bar{\tau}^2$ the average posterior rate variance at the forecast points. The two contributions scale differently in δ_f, which is what lets us separate them. Naive averaging like $\sigma^2_{\mathrm{session}} \approx \mathrm{mean}_f(e_f^2/\delta_f)$ would not separate them and would bias the estimate upward by a $\delta_f\,\bar{\tau}^2$ contamination.

Joint regression. Fit both unknowns by OLS of e_f^2 on $[\delta_f,\, \delta_f^2]$ (no intercept):

$$(\hat{a}, \hat{b}) = \arg\min_{a,b} \sum_f \big(e_f^2 - a\,\delta_f - b\,\delta_f^2\big)^2, \qquad \sigma^2_{\text{session}} := \max(\hat{a},\, 10^{-6}).$$

The coefficient $\hat{a}$ on the linear term is the per-hour path-noise variance: that is the quantity we want.

Note that this regression treats $\bar{\tau}^2$ as constant across forecast points, while in reality $\tau^2_{\text{post}}(t_f)$ varies (it depends on how many recent snapshots existed at t_f). If $\tau^2_{\text{post}}(t_f)$ correlates with δ_f — and it usually does, since δ_f is anti-correlated with elapsed time in the session — the missing covariance leaks into $\hat{a}$. A tighter alternative is to subtract the known $\delta_f^2\,\tau^2_{\text{post}}(t_f)$ piece as an offset and regress the residual on δ_f alone; we keep the joint regression for simplicity and absorb the bias into the daily refit.

Why $\hat{b}$ is discarded. Conceptually $\hat{b}$ estimates the same quantity as τ^2_{post} in §5 (the rate-estimation variance), just as a historical average instead of a fresh per-forecast value. The per-forecast version in §5 uses today's actual snapshots, so it is strictly more informative; $\hat{b}$ is redundant.

The $b\,\delta_f^2$ term still has to be *present in the regression*, though. Without it, the $\delta_f^2\,\bar{\tau}^2$ piece of e_f^2 has nowhere to go but $\hat{a}$, biasing it upward. So b is included as a nuisance parameter (analogous to adjusting for a confounder in a regression: we control for its contribution without caring about its fitted value), and discarded once $\hat{a}$ is clean.

Floor and refit. The floor on $\hat{a}$ guards against negative estimates from small samples (OLS does not enforce sign). Refit daily.

8 Threshold ETA

For threshold C_{thr} with $u_{\text{now}} < C_{\text{thr}} \leq 1$, the first-passage time is $T^* = \inf\{t > t_{\text{now}} : u(t) \geq C_{\text{thr}}\}$, with $T^* = \infty$ if no crossing in $[t_{\text{now}},\, T_{\text{reset}}]$.

Deterministic ETA (point estimate). Setting $W \equiv 0$ and $r = \hat{r}_{\text{post}}$:

$$\tilde{T^*} = t_{\text{now}} + \frac{C_{\text{thr}} - u_{\text{now}}}{\hat{r}_{\text{post}}},$$

defined when $\hat{r}_{\text{post}} > 0$ and $\tilde{T^*} \leq T_{\text{reset}}$; else `nil`. Note that $\tilde{T^*}$ is **not** the median of T^* under the full stochastic model. Two effects pull in opposite directions:

- *BM skew (pulls median down).* For BM with fixed drift $\mu > 0$ hitting $a > 0$, $T^* \sim \mathrm{IG}(a/\mu,\, a^2/\sigma^2)$ with mean a/μ and right skew, so $\mathrm{median}(T^* \mid r{=}\mu) < a/\mu$.
- *Rate uncertainty (pulls median up).* Mixing over $r \sim \mathcal{N}(\hat{r}_{\text{post}},\, \tau^2_{\text{post}})$ inflates first-passage times asymmetrically: the map $r \mapsto a/r$ is convex on $r > 0$ and steepens sharply as $r \to 0^+$, so the right tail of T^* stretches far more than the left tail compresses. Any mass at $r \leq 0$ contributes additional atoms at $T^* = \infty$. Both effects pull the median of T^* above $\tilde{T^*}$.

Which effect wins depends on the regime, so $\mathrm{median}(T^*)$ can sit on either side of $\tilde{T^*}$. The MC percentiles below are the authoritative summary; $\tilde{T^*}$ is a cheap deterministic anchor, not an estimator of the median.

CI via Monte Carlo. The first-passage time of a Brownian motion with random Gaussian drift has no tractable closed form. Simulate $K = 500$ trajectories:

```
for k in 1..K:
    r_k    = N(r_hat_post, tau_post^2)     # may be negative; do NOT clip
    u_k[0] = u_now
    for j in 1..M, dt = step size of jth bucket:
        u_k[j] = u_k[j-1] + r_k * dt + N(0, sigma_session^2 * dt)
    T*_k = linear interpolation of first j with u_k[j] >= C_thr
         = infinity if none
```

Negative r_k draws are kept as-is: those trajectories drift downward and typically yield $T_k^* = \infty$, which is the correct outcome under the model. Clipping to zero would bias crossings upward and contradict §5.

Reported summaries. Let p_∞ be the fraction of trajectories with $T_k^* = \infty$. Define $\tilde{T^*}_{\mathrm{MC}} = \mathrm{median}(T_k^*)$ over the **full** sample (treating ∞ as larger than any finite value):

- If $p_\infty \geq 0.5$: the true median is ∞. Report ETA as `nil`.
- If $p_\infty < 0.5$ but $p_\infty \geq 0.1$: the median is finite but the 90th percentile is ∞. Report median, lower bound = 10th percentile of finite T_k^*, upper bound = `nil` (open-ended).
- If $p_\infty < 0.1$: report median and full 80% CI from the 10th and 90th percentiles of finite T_k^*.

RNG seed derived from $(t_{\mathrm{now}}, T_{\mathrm{reset}}, u_{\mathrm{now}}, \hat{r}_{\mathrm{post}}, \tau^2_{\mathrm{post}}, \sigma^2_{\mathrm{session}})$ for test reproducibility.

9 Confidence tag

Summarises the effective amount of evidence:

$$N_{\mathrm{eff}} = \min\left(n,\ \frac{\tau_0^2}{\mathrm{SE}^2_{\mathrm{OLS}}} + |\mathcal{S}|\right).$$

Tag	Threshold
High	$N_{\mathrm{eff}} \geq 50$
Medium	$15 \leq N_{\mathrm{eff}} < 50$
Low	$N_{\mathrm{eff}} < 15$

Thresholds are heuristics; revisit after seeing real values.

10 Edge cases

Case	Handling
$n < 3$ snapshots in $\mathcal{R}$	Use prior alone: $\hat{r}_{\mathrm{post}} = \mu_0$, $\tau^2_{\mathrm{post}} = \tau^2_0$
Prior empty ($|\mathcal{S}| < 2$)	Suppress forecast; UI shows "collecting data"
Reset detected in $\mathcal{R}$ ($u_{i+1} < u_i$)	Drop pre-reset snapshots; refit
$C_{\mathrm{thr}} \leq u_{\mathrm{now}}$	Threshold already crossed; return $T^* = t_{\mathrm{now}}$
$\hat{r}_{\mathrm{post}} \leq 0$	Deterministic ETA undefined; rely on MC summary
$F \notin [0, 1]$	Clip for display; retain unclipped for downstream

11 Worked example

5-hour session window: $t_{\text{start}} = 11{:}00$, $T_{\text{reset}} = 16{:}00$, $t_{\text{now}} = 13{:}00$, $u_{\text{now}} = 0.30$, so $\Delta t_{\text{rem}} = 3\,\text{h}$.

Last four snapshots ($\tau_{\text{recent}} = 30\,\text{min}$):

i	t_i (h from 12:30)	u_i
1	0.000	0.270
2	0.167	0.280
3	0.333	0.295
4	0.500	0.300

OLS: $\hat{r}_{\text{OLS}} \approx 0.0630\,\text{h}^{-1}$, $\text{SE}^2_{\text{OLS}} \approx 6.30 \times 10^{-5}$.
Prior (suppose, from history, after the §5 noise correction): $\mu_0 = 0.080\,\text{h}^{-1}$, $\tau_0^2 = 3.6 \times 10^{-3}$.
Posterior: $\tau^2_{\text{post}} \approx 6.19 \times 10^{-5}$, $\hat{r}_{\text{post}} \approx 0.0633\,\text{h}^{-1}$ (data dominates).
Calibration: $\sigma^2_{\text{session}} = 2.5 \times 10^{-3}\,\text{h}^{-1}$.
Forecast:

$$F = 0.30 + 0.0633 \cdot 3 \approx 0.490,$$

$$\sigma_F^2 \approx 9 \cdot 6.19 \times 10^{-5} + 3 \cdot 2.5 \times 10^{-3} \approx 5.6 \times 10^{-4} + 7.5 \times 10^{-3} \approx 8.1 \times 10^{-3},$$

$$\sigma_F \approx 0.090.$$

80% CI: $[0.490 - 1.282 \cdot 0.090,\ 0.490 + 1.282 \cdot 0.090] \approx [0.375,\ 0.605]$. Display: "Projected 49% by reset (80% CI: 38%–60%)".

Path noise dominates rate uncertainty by 13×. That is: we know the current slope well; the spread is "you might pivot."

ETA to 100%: gap is 0.70, deterministic crossing at $13{:}00 + 0.70/0.0633 \approx 24{:}04$, far beyond reset. Returned as `nil`. The MC confirms by producing infinite crossings for almost all trajectories.

12 Limitations and upgrade paths

- **Heavy-tailed pivots.** Real per-hour increments are heavier-tailed than Gaussian. The 80% CI undercovers in the tails. Cheap fix when calibration data is plentiful: replace $z_{0.9}$ with an empirical quantile of standardized residuals.
- **Constant-rate assumption.** A mode shift within the window (deep work → meetings) breaks (A1). The recency window τ_{recent} absorbs slow drift but not abrupt shifts. The principled upgrade is a Kalman filter where r is a slowly-varying latent process.
- **Global $\sigma^2_{\text{session}}$.** Pivot variance may depend on day-of-week or project type. Stratified calibration is the upgrade, but requires more sessions per stratum.